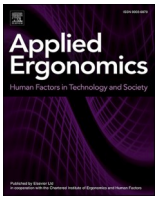
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Traditional versus dynamic sitting: Lumbar spine kinematics and pain during computer work and activity guided tasks

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ABSTRACT

Dynamic sitting may mitigate low back pain during prolonged seated work. The current study compared pelvis and lumbar spine kinematics, pain, and work productivity, in traditional and dynamic sitting. Sixteen participants completed three 20-min blocks of computer work and activity guided tasks in a traditional office chair or backless and multiaxial rotating seat pan while kinematics were measured from accelerometers on the low back. Pain ratings were recorded on a visual analogue scale every 10 min. Similar pelvis and lumbar kinematics emerged when performing computer work in traditional and dynamic sitting. Pelvis and lumbar sagittal and frontal plane shifts and fidgets were largest for dynamic sitting in the activity guided tasks. Buttocks pain was higher in dynamic sitting, but low back pain and work productivity were unaffected. Dynamic sitting increased spine movement during activity guided tasks, without negatively impacting lumbar kinematics, low back pain, or productivity during seated computer work.

1. Introduction

An association between prolonged sitting and low back pain has been observed in both epidemiological (da Silva et al., 2019; Gupta et al., 2015; Hanna et al., 2019; Levangie, 1999; Lis et al., 2007; Mendelek et al., 2011; Park et al., 2018; Thorbjörnsson et al., 2000; Van Vuuren et al., 2005) and laboratory research (De Carvalho et al., 2020; O'Keeffe et al., 2013; O'Sullivan et al., 2013; O'Sullivan et al., 2006; Schinkel-Ivy et al., 2013; Sheahan et al., 2016; Vergara and Page, 2002; Williams et al., 1991). Sustained lumbar flexion and decreases in spine movement relative to other activities of daily living are potential mechanisms for sitting-induced low back pain. Sitting situates the spine into flexion, typically between 30% and 80% of the maximum voluntarily lumbar flexion angle (%MaxFlex) (Beach et al., 2005; Callaghan and McGill, 2001; Dunk and Callaghan, 2005; Greene et al., 2019). Lumbar flexion may lead to the perception of pain as it alters the spine loading environment such that the posterior passive tissues of the spine are engaged (Callaghan and McGill, 2001) and a greater proportion of the compressive load is supported by the intervertebral disc (Adams et al., 1994; Adams and Hutton, 1985; 1980; Hedman and Fernie, 1997; Nachemson, 1966). A relative decrease in spine movement while sitting may also contribute to the perception of pain as spine movement is necessary to redistribute loads, fluid, and nutrients throughout the joint (Adams and Hutton, 1985; Huang et al., 2014; Nachemson, 1966; Wilder et al., 1988; Wilke et al., 1999). Specifically, those with low back pain often exhibit larger and/or more frequent micromovements, indicating that increases in spine movement may be a mechanism to alleviate or combat low back pain (De Carvalho and Callaghan, 2023a; Dunk and Callaghan, 2010; Vergara and Page, 2002). Taken together, interventions which decrease lumbar flexion and promote movement of the lumbar spine in sitting may assist in reducing the risk of low back pain with sitting.

Multiple interventions targeting reduced spine flexion and/or increased spine movement in prolonged sitting have been proposed. Recent popular interventions include whole-body movement or hybrid seating such as taking standing or walking breaks to reduce the time spent sitting. A systematic review noted decreases in worker pain when utilizing a sit-stand workstation compared to sitting only (Karakolis and Callaghan, 2014). Unfortunately, when sit-stand workstations are implemented in the workplace, individuals often do not stand enough, such that sitting time is not significantly reduced (Graves et al., 2015; Pronk et al., 2012; Riddell and Callaghan, 2021; Robertson et al., 2013; Wilks et al., 2006). Further, taking a break from sitting (e.g., 2 min of walking every 40 min) only temporarily relieves sitting-induced low back pain (De Carvalho and Callaghan, 2023a). As an alternative to sit-stand rotation, hybrid sit-stand chairs have been proposed to avoid

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end-range spine flexion in sitting (Antle et al., 2015; Chester et al., 2002; Ebara et al., 2008; Kingston et al., 2016; Le and Marras, 2016; Sartika and Dawal, 2010). However, recommendations for appropriately positioning the spine in these hybrid chairs are more recent (Buchman-Pearle et al., 2023; Noguchi et al., 2019) and there is little evidence that simply opening up the trunk-thigh angle in hybrid chairs will elicit less spine flexion than a traditional 90-90-90 sitting posture. Unlike hybrid chairs which primarily aim to decrease spine flexion in sitting, 'dynamic' or 'active' chairs aim to decrease spine flexion while also promoting movement. For instance, dynamic chairs use moveable or deformable seat pans which could tilt forward to decrease pelvic tilt and tilt side-to-side to increase lateral pelvis movement. Many dynamic chairs variations exist, such as stability balls, traditional office chairs with a split seat pan, and stools with a multiaxial rotating seat pan. In a multiaxial rotating seat pan without a back support (like that investigated currently) pelvis, hip, and trunk-thigh angles and trunk muscle activation were not different from a traditional office chair, but buttocks pain was higher (Léger et al., 2022). Spine flexion and fidget frequency or amplitude were also not different between dynamic and traditional sitting, but low back pain remained lower in dynamic sitting (Frey et al., 2021). Dynamic sitting on a different rotating seat pan also decreased mean lumbar flexion by ~25 %MaxFlex with no differences in lumbar flexion variability or pain ratings compared to a traditional office chair with the seatback removed (O'Sullivan et al., 2012). Evaluations comparing the different types of dynamic chairs are limited, but they generally demonstrate similar responses across chair types. Trunk muscle activation was not different between a rotating and split seat pan but the rotating seat pan tended to promote greater pelvis range of motion in the sagittal and frontal plane (Léger et al., 2022). Trunk muscle activation was also similar between a stability ball and rotating seat pan when completing core exercises (Holmes et al., 2015). Additionally, while buttocks pain increased on both types of dynamic chairs, buttocks pain was higher on the split seat pan design (Léger et al., 2022). Collectively, it appears that dynamic sitting, facilitated by a rotating seat pan, has the potential to alter lumbar posture and movement, but the results depend on the chair being assessed. Additionally, despite that some rotating seat pans have the capability to promote movement in both the sagittal and frontal plane (i.e., seat pan tilts forward and backward and side-to-side), studies have not typically assessed kinematic changes in the frontal plane. While the opportunity for lateral tilt of the seat pan may increase spine movement, asymmetry in spine postures could also increase the risk of pain (Marras et al., 1997). However, lateral spine and pelvis kinematics have not been investigated in dynamic sitting contexts.

The purpose of this investigation was to evaluate how dynamic sitting altered pelvis and lumbar spine kinematics in the sagittal and frontal plane and perceived pain. Dynamic sitting while performing computer work and activity guided was facilitated by using a chair with a multiaxial rotating seat pan and was compared to traditional sitting. Seated activity guided tasks can provide an opportunity to train individuals on how to use a dynamic chair and promote additional movement at the workplace. While active breaks usually involve a more extended period away from the workstation (e.g., 30-min walk), activity guided tasks performed at the workstation could provide supplemental micro or short breaks (e.g., performed as a break or in conjunction with non-visual tasks like a phone call). We hypothesized that compared to traditional sitting, dynamic sitting would demonstrate similar pelvis and lumbar postures and movement in the sagittal plane (Frey et al., 2021); however, increases in frontal plane pelvis and lumbar postures and movement were expected due to the increased lateral mobility of the seat pan compared to a traditional office chair. Additionally, it was hypothesized that buttocks pain would modestly increase during dynamic sitting (Léger et al., 2022), but that low back pain would not be different (Frey et al., 2021; O'Sullivan et al., 2012). Secondary objectives of this study were to assess differences in trunk-thigh and hip flexion postures between the seats, weight distribution on the lower limbs versus under the seat, and work productivity.

2. Methods

2.1. Sample population

Eight males (age = 25.5 ± 2.5 years; height = 1.82 ± 0.06 m; mass = 80.7 ± 7.1 kg; body mass index = 24.2 ± 1.3 kg/m^2) and eight females (age = 23.3 ± 1.2 years; height = 1.73 ± 0.08 m; mass = 69.5 ± 15.0 kg; body mass index = 23.0 ± 4.1 kg/m^2) participated in this study. A sample size calculation indicated that 16 participants were required for a medium to large effect size with an alpha of 0.05 and beta of 0.8 (G*Power, Version 3.1.97, Kiel, Germany). Individuals were excluded if they reported having received medical attention or taken time off work for longer than three days for low back pain or injury within the past year or had a physical disability, particularly those which would affect mobility or dexterity. The University Research Ethics Board approved all experimental procedures, and each participant signed informed consent documentation.

2.2. Experimental protocol

Testing occurred in a single laboratory session where individuals completed three 20-min blocks of: a) seated computer work in a control condition of a traditional office chair (Traditional; Herman Millar, Aeron, Zeeland, Michigan, USA), b) seated computer work on a dynamic chair (Dynamic; QoR360°, Ariel 1.0, Burlington, Vermont, USA), and c) activity guided tasks on the same dynamic chair (Activity; Fig. 1). To facilitate dynamic sitting, participants sat on a chair with a multiaxial rotating seat pan which enabled a maximum of ~10° tilt to the left and right, ~15° forwards, and ~5° backwards. Order of blocks a) and b) were randomized with block c) completed last to avoid any potential confounding effects of stretching soft tissues during the activity guided tasks. Between blocks, individuals were permitted to stand up to facilitate switching between the chairs. In blocks a) and b) individuals performed a standardized data entry task on the computer for the full 20 min. The data entry task involved the transfer of product, client, and company information from invoices with a consistent number of entries into a custom developed graphic user interface (i.e., required mousing and typing). In block c) individuals performed two activities from the QOR360 FitterSitter© application (QOR360 FitterSitter, Version 1.3.4, Burlington, Vermont, USA). The FitterSitter© application determines the inclination of the seat pan from the tilt of a phone's accelerometer



Fig. 1. The chairs used for computer work in traditional sitting (left) and computer work and activity guided tasks in dynamic sitting (right).

which is coupled to a chair's seat pan. Individuals are guided through the activities using real-time feedback of the chair movement from the phone shared to a computer. In this study the phone was attached to the back of the seat pan and the chair's movement was shared to the monitor on the desk. The activities were designed to be performed on a multi-axial rotating seat pan, as they were controlled through motion of the pelvis in the sagittal and/or frontal plane. Participants played Pong for the first 5-min, followed by 15-min of Free Play. Pong was the traditional video game requiring the participant to move a paddle left and right to return a moving ball (facilitated motion predominantly in the frontal plane). Free Play involved target tracking in 2D to maintain a ball within a target circle (facilitated motion in the sagittal and frontal plane). Throughout the activities, individuals were instructed to rest their hands on the desk in front of them and achieve the desired motions by primarily moving their pelvis, with their feet situated on the force plates.

Prior to beginning each block, the chair and workstation parameters were adjusted to each individual's anthropometry. The height of the dynamic chair was set to facilitate a trunk-thigh angle between 105° and 125°, as measured by a goniometer. The traditional office chair was set to facilitate a traditional starting position of 90° trunk-thigh, knee, and ankle angle (90-90-90). The desk height facilitated a horizontal forearm position with the individuals' hands on the keyboard and the top of the monitor was aligned with the individual's horizontal line-of-sight (CSA Group, 2017). To determine the functional range of motion for the dynamic chair for each participant and facilitate calibration of the accelerometer in the phone, and hence the inclination of the seat pan, for the FitterSitter© application, individuals were instructed to tilt the seat pan maximally to the left, right, forward, and backwards while seated in the dynamic chair. Individuals performed all three blocks on three embedded force plates (AMTI, Watertown, MA, USA), in which one force plate was under the chair (Model BP900) and one force plate was under each foot (Model MC12 and ORE6-7). Force data was collected at 250 Hz.

2.3. Instrumentation

Triaxial accelerometers (ADXL 335, Analogue Devices, Norwood, MA) were used as inclinometers to measure pelvis and lumbar spine angles. The accelerometers were affixed to the skin overlaying the first sacral (S1) and lumbar vertebrae (L1). Calibration trials, namely upright standing, maximum standing forward flexion, and maximum lateral bending to the right and left, were performed to normalize the flexion-extension and lateral bending angles with respect to the participants' range of motion. Accelerometer data were sampled at 250 Hz.

Rigid marker clusters, each containing four infrared markers (Northern Digital Inc., Waterloo, ON) measured trunk-thigh and hip angles. Marker clusters were secured to the skin overlying the sternum and the lateral aspect of the thigh. A separate rigid body cluster was temporarily secured directly on top of the S1 accelerometer during the calibration trials only. A static upright standing calibration trial was recorded, and the proximal and distal ends of segments were digitized and tracked relative to the segments of interest. Functional joint trials were bilaterally performed to identify the position of the hip joint center (Camomilla et al., 2006; Schwartz and Rozumalski, 2005). The hip joint centers were tracked with the respective thigh segment during the seated conditions to facilitate construction of the pelvis segment (Zehr et al., 2021). Marker position data were sampled at 25 Hz.

Individuals completed pain ratings bilaterally for the upper back, low back, buttocks, and thighs using a 100 mm visual analogue scale (VAS) (Tennant and Lab, 2023). A rating of 0 mm indicated "No Pain" and a rating of 100 mm indicated "Worst Pain Imaginable". In each 20-min block, pain ratings were obtained at: 0, 10, and 20 min.

2.4. Data processing

All kinematic data were padded (Howarth and Callaghan, 2009) and then smoothed using a dual-pass second order low-pass Butterworth filter with an effective cutoff frequency of 1 Hz (Buchman-Pearle et al., 2021; Dunk and Callaghan, 2010; Greene et al., 2019). Accelerometer voltage data were calibrated to gravitational acceleration (g) and converted to degrees using standard four-quadrant trigonometric equations. Pelvis anterior-posterior tilt and left and right obliquity were derived from the inclination of the S1 accelerometer. Lumbar flexion-extension and left and right lateral bending were derived from the relative inclination between the S1 and L1 accelerometers. All angles were expressed relative to the angles from the upright standing trial and the absolute of the lateral angles were calculated. Further, lumbar angles were normalized to maximum flexion and lateral bending range of motion (i.e., mean of the absolute angle from the left and right maximum bending trials).

Pelvis anterior-posterior tilt and obliquity and lumbar flexion-extension and lateral bending angles were assessed by an amplitude probability distribution function (APDF) for each 20-min block (Haoberg and Jonsson, 1975). Dependent variables for statistical analyses were angles at the 10%, 50%, and 90% of the APDFs for pelvis anterior-posterior tilt and obliquity and lumbar spine flexion-extension and lateral bending.

Movement of the lumbar spine in the sagittal and frontal planes were evaluated by calculating the shift and fidget frequency and amplitude using modified parameters presented by Duarte and Zatsiorsky (1999), which were later adapted by Dunk and Callaghan (2010) to measure micromovements of the lumbar spine in sitting. Individuals with sitting-aggravated low back pain demonstrated more frequent shifts and larger shifts and fidgets, potentially as mechanism to alleviate their pain (Dunk and Callaghan, 2010). Shifts were defined when the difference in the average lumbar angle of two 15-s sliding windows, separated by 3-s, exceeded 5°. The shift amplitude was calculated by the average difference in the angle of two consecutive windows throughout the whole shift. Fidgets were defined when the lumbar angle at a data frame exceeded 3 standard deviations of the 60-s sliding window centered on that frame, for no longer than 3-s. The fidget amplitude was calculated by the average difference in the angles throughout the whole fidget and the average of the 60-s window. The total number of shifts and fidgets and the mean amplitude of the shifts and fidgets in the sagittal and frontal plane throughout the whole block were calculated. Lumbar flexion-extension and lateral bending shift and fidget frequency (per 20 min) and average amplitude were input into statistical analyses.

Data acquired from the upright standing trial were used to construct segment-fixed local coordinate systems for the thorax, pelvis, and thighs. Given the obstructed line-of-sight to the pelvis cluster during the seated conditions (i.e., due to the individual's close interface with the seatback and seat pan), a three-dimensional accelerometer-based pelvis was calculated using the angular deviations of the S1 accelerometer and a three-dimensional vector rotation approach (Zehr et al., 2021). Trunk-thigh (thigh relative thorax) and hip (thigh relative to accelerometer-based pelvis) were calculated using a Z-X-Y Cardan sequence (Wu et al., 2002). Average trunk-thigh and right and left hip flexion angles were input into statistical analyses.

Force data were padded (Howarth and Callaghan, 2009) and then smoothed using a dual-pass second order low-pass Butterworth filter with an effective cutoff frequency of 10 Hz (Noguchi et al., 2019) and expressed as a percentage of body weight. The force in the vertical and anterior-posterior direction were computed separately to indicate if individuals tended to forward brace on their legs.

From the VAS, pain ratings for each body region were expressed

relative to the values collected at the start of the data collection (i.e., subtract baseline) and an average value was calculated for the left and right side.

Productivity was assessed for the data entry task completed in blocks a) and b). Productivity was evaluated based on accuracy (percentage of correct entries) and number of entries per minute spent typing.

2.5. Statistical analysis

Statistical analyses were performed on the trunk-thigh, hip, pelvis, and lumbar spine kinematics, force, pain, and productivity variables to determine the effects of *block* (within-subject factor: Traditional, Dynamic, and/or Activity) and *sex* (between-participant factor: male or female). Sex was investigated as a factor because differences in seated kinematics emerged between males and females (De Carvalho and Callaghan, 2023a; Dunk and Callaghan, 2005; Gregory et al., 2006). When applicable, the effect of *time* (within-subject factor: pain = 3 instances in each block) and *side* (within-subject factor: hip = left and right) were also evaluated. Specifically, mixed model analyses of variance (ANOVA) were performed at an alpha level of 0.05. When significance was achieved, post hoc pairwise comparisons were performed with a Bonferroni correction. All statistical tests were completed in SPSS (Version 28.0.1, IBM Corporation, Somers, New York, USA).

3. Results

3.1. Trunk-thigh and hip kinematics

A main effect of *block* was observed for the trunk-thigh and hip flexion angle ($p \leq 0.001$). The average trunk-thigh angle was smaller during traditional sitting, as expected given the different positioning guidelines, compared to dynamic sitting for both the computer work and activity guided tasks ($p_{adj} \leq 0.012$; Fig. 2A). The average hip flexion angle was also smaller during traditional sitting compared to dynamic sitting in both tasks ($p_{adj} \leq 0.047$; Fig. 2B). No significant effects of *sex* were observed for hip or trunk-thigh angles ($p \geq 0.358$).

3.2. Pelvis and lumbar spine postures

For anterior-posterior pelvic tilt, no significant main effects or interactions for *sex* or *block* were observed at 10%, 50%, and 90% of the APDF ($p \geq 0.060$). However, throughout the blocks, the pelvis was typically less posteriorly tilted in dynamic sitting during computer work (26.2°–29.3°) and the activity guided tasks (26.4°–35.2°) compared to traditional sitting (32.4°–35.2°; Fig. 3A).

For lumbar flexion-extension angles, no significant main effects or interactions for *block* or *sex* were observed at 10%, 50%, and 90% of the APDF ($p \geq 0.109$; Fig. 3B). Throughout the block, lumbar flexion remained consistent for computer work in traditional sitting (62.3%) and dynamic sitting (62.4%), and the activity guided tasks (68.5%) (Fig. 3B).

For pelvic obliquity, a main effect of *block* was observed at 10% and 90% of the APDF ($p \leq 0.048$). There were no significant differences in pelvic obliquity between traditional and dynamic sitting during the computer work ($p_{adj} \geq 0.700$). At 10%, pelvic obliquity was marginally smaller during the activity guided tasks (1.4°) than the computer work during dynamic sitting (2.7°; $p_{adj} = 0.002$). At 90%, pelvic obliquity was larger in the activity guided tasks than during computer work performed on either chair ($p_{adj} \leq 0.001$; Fig. 4A). No significant effects of *sex* were observed for pelvic obliquity at any percent of the APDF ($p \geq 0.341$).

For lumbar lateral bending, no significant effects of *block* were observed at 10% or 50% of the APDF ($p \geq 0.104$), but a main effect of *block* was observed at 90% ($p \leq 0.001$). There were no significant differences in lateral bending between traditional and dynamic sitting during the computer work ($p_{adj} = 1.000$), but lateral bending was larger during the activity guided tasks than computer work on either chair ($p_{adj} \leq 0.001$; Fig. 4B). No significant effects of *sex* were observed for lateral bending at any percent of the APDF ($p \geq 0.090$).

3.3. Lumbar spine shifts and fidgets

There were very limited lumbar shifts present across participants. Of the 16 individuals, flexion-extension shifts were identified for 6 (Traditional), 2 (Dynamic), and 9 (Activity) individuals and lateral bending shifts were identified for 2 (Traditional), 0 (Dynamic), and 14 (Activity) individuals. There were no significant main effects or interactions for *block* or *sex* on lumbar flexion-extension shift frequency ($p \geq 0.097$) or shift amplitude ($p \geq 0.122$; Table 1).

A main effect of *block* was observed for lumbar lateral bending shift frequency ($p = 0.050$). Yet pairwise comparisons revealed no significant

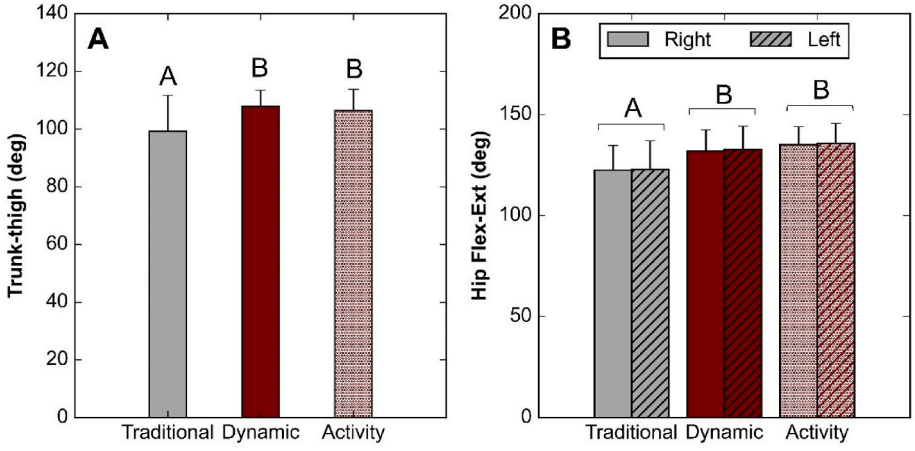


Fig. 2. Mean and (standard deviation) of the trunk-thigh angle (A) and hip flexion (B) angle in each 20-min block. Bars with different capital letters indicate significant differences between *block*.

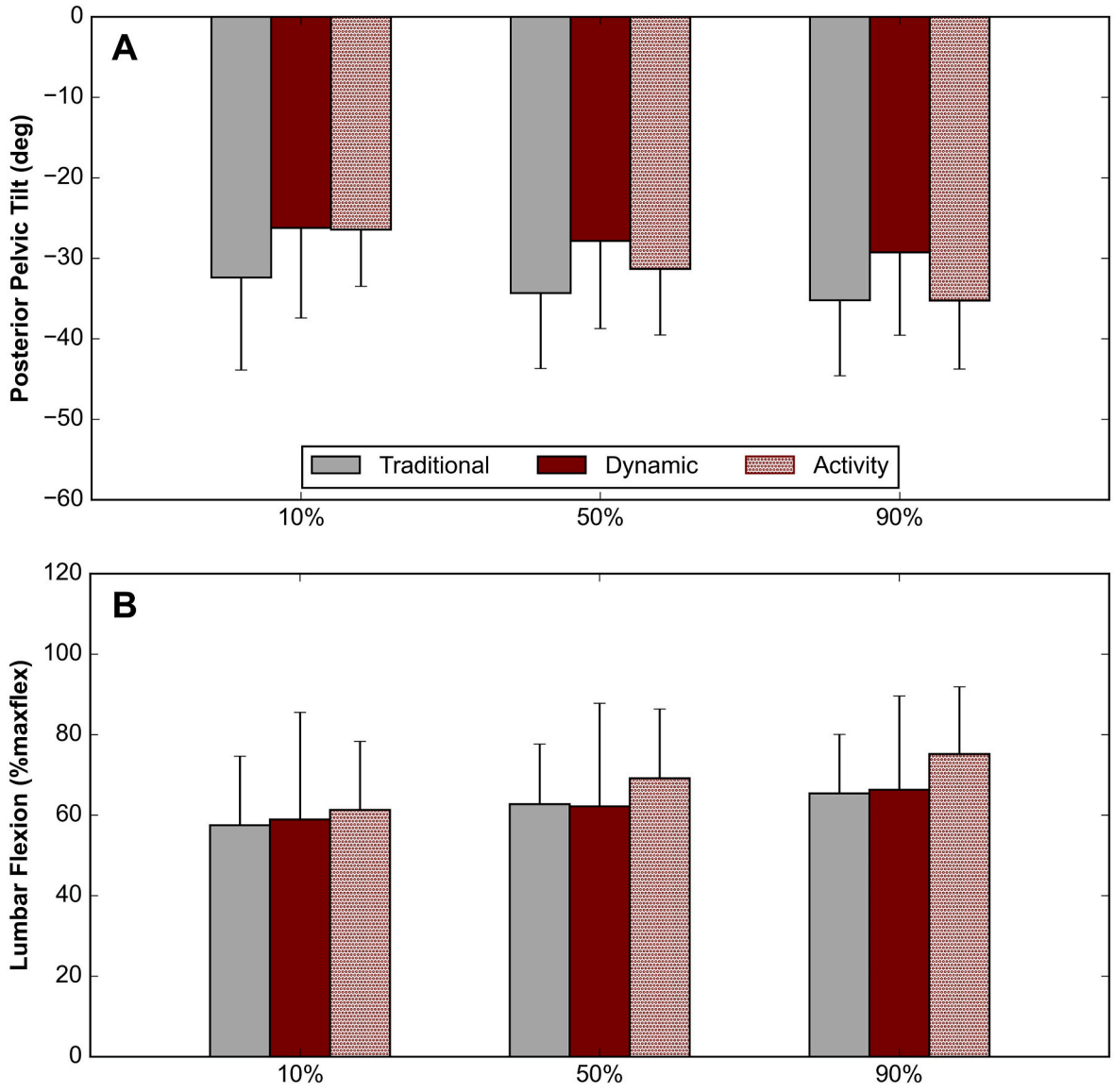


Fig. 3. Mean and (standard deviation) of pelvis anterior posterior tilt (A) and lumbar flexion-extension (B) at 10%, 50%, and 90% of the amplitude probability distribution function (APDF).

differences ($p_{adj} \geq 0.195$; Table 1). A main effect of *block* was observed for lumbar lateral bending shift amplitude ($p \leq 0.001$); however, there was an extremely low incidence of shifts during the computer work compared to the activity guided tasks. Nevertheless, of the shifts observed, the shifts were larger in the activity guided tasks ($p_{adj} \leq 0.003$; Table 1). No significant effects of *sex* were observed for lateral bending shift frequency or amplitude ($p \geq 0.195$).

There were no significant main effects or interactions for *block* or *sex* on lumbar flexion fidget frequency ($p \geq 0.561$; Table 1). There was a significant *block* × *sex* interaction for the lumbar flexion fidget amplitude ($p = 0.024$). Larger fidgets were consistently observed in the activity guided tasks compared to the computer work (Table 1), but the increase in the fidget amplitude was larger for females (Traditional = 3.4 ± 1.5%, Dynamic = 5.4 ± 4.3%, Activity = 15.2 ± 2.8%) than males (Traditional = 5.2 ± 3.0%, Dynamic = 5.5 ± 2.9%, Activity = 11.6 ± 5.4%; $p_{adj} \leq 0.006$).

A main effect of *block* was observed for lumbar lateral bending fidget frequency ($p = 0.001$). Pairwise comparisons revealed that individuals performed fewer fidgets during the activity guided tasks ($p_{adj} \leq 0.012$; Table 1). A main effect of *block* was also observed for lumbar lateral bending fidget amplitude ($p \leq 0.001$). Significant differences were observed between all blocks such that traditional sitting demonstrated the smallest fidgets and activity guided tasks demonstrated the largest ($p \leq 0.001$; Table 1). No significant effects of *sex* were observed for lateral bending fidget frequency or amplitude ($p \geq 0.228$).

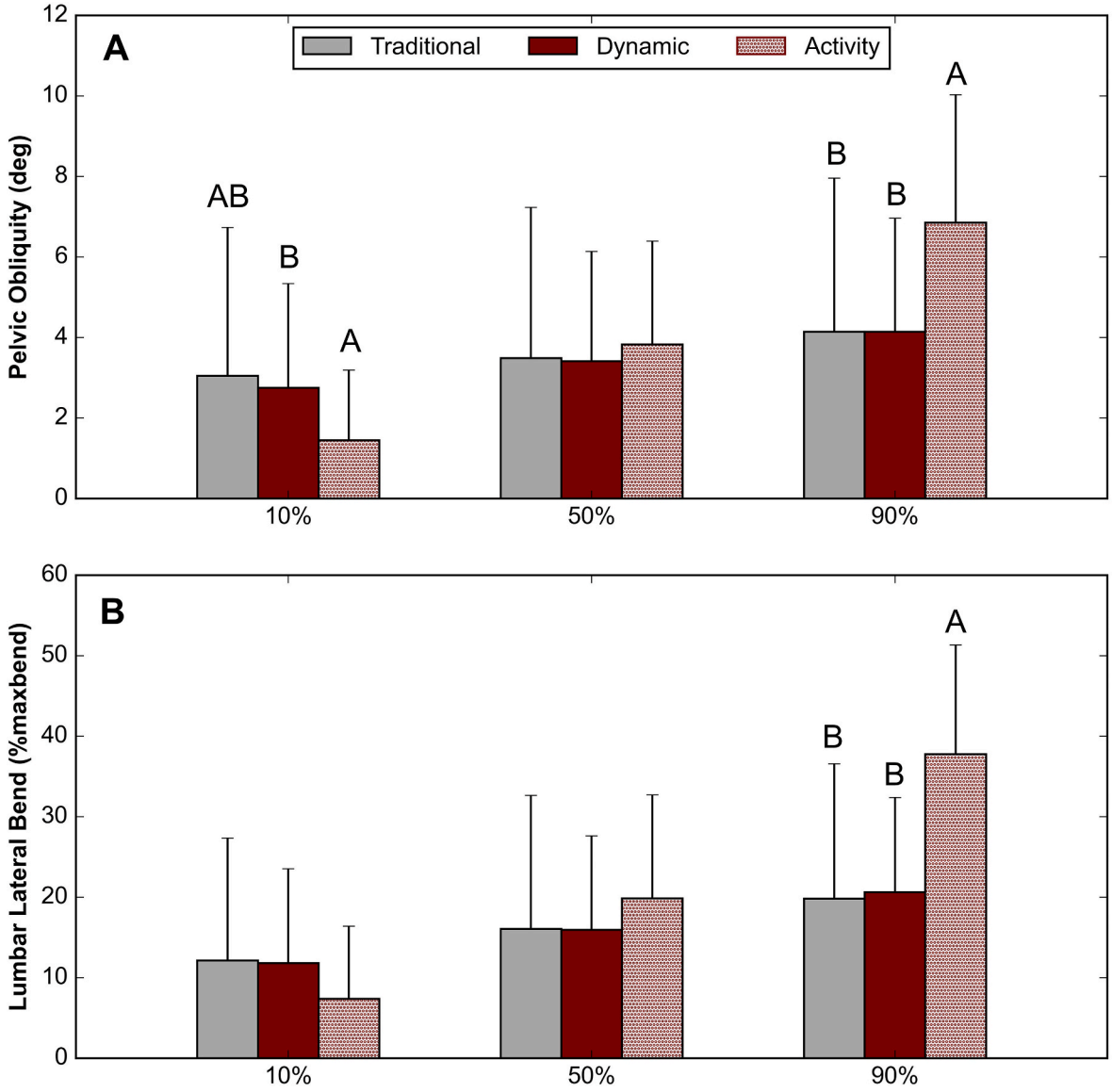


Fig. 4. Mean and (standard deviation) of absolute pelvis obliquity (A) and lumbar lateral bending (B) at 10%, 50%, and 90% of the amplitude probability distribution function (APDF). Bars with different capital letters indicate significant differences between *block*.

Table 1
Mean and (standard deviation) of lumbar flexion and lateral bending shift and fidget frequency in each 20-min block. Values with different capital letters indicate significant differences between *block*.

	Frequency (per 20 min)			Amplitude (%Max)		
	Traditional	Dynamic	Activity	Traditional	Dynamic	Activity
Shift						
Sagittal	1.5 (3.5)	1.5 (4.4)	4.0 (5.4)	5.0 (6.8)	2.3 (5.1)	6.2 (5.8)
Frontal	0.2 (0.5)	0.0 (0.0)	6.3 (11.8)	3.2 (8.7)^{A}	0.0 (0.0)^{A}	22.9 (9.8)^{B}
Fidget						
Sagittal	12.9 (7.0)	11.9 (5.8)	13.0 (6.0)	4.3 (2.5)	5.4 (3.6)	13.4 (4.5)
Frontal	14.8 (10.9)^{A}	12.5 (8.0)^{A}	5.1 (3.9)^{B}	4.7 (4.4)^{A}	8.1 (3.7)^{B}	32.0 (14.5)^{C}

3.4. Ground reaction force

There was a main effect of *block* for the anterior-posterior and vertical directions ($p \leq 0.002$). The anterior-posterior force was significantly larger during the activity guided tasks ($p_{adj} \leq 0.002$; Table 2). The vertical force under the feet was significantly larger in both dynamic sitting conditions than traditional sitting ($p_{adj} < 0.001$; Table 2).

3.5. Pain

There were no significant main effects or interactions for *block*, *sex*, or *time* on pain ratings for the upper back ($p \geq 0.074$; Fig. 5A) or thighs ($p \geq 0.071$; Fig. 5D). No significant main effects or interactions for *block* or *sex* were observed for pain ratings for the low back ($p \geq 0.116$), but there was a main effect of *time* ($p = 0.007$). Low back pain was highest at

Table 2
Mean and (standard deviation) for the force under the feet, normalized to percent bodyweight (%BW), in each 20-min block. Values with different capital letters indicate significant differences between *block*.

Direction	Traditional	Dynamic	Activity
Anterior (+)-posterior (%BW)	1.4 (0.8)^{A}	1.7 (1.2)^{A}	3.2 (1.1)^{B}
Vertical (%BW)	13.5 (2.4)^{A}	22.7 (1.6)^{B}	21.2 (2.6)^{B}

20-min and significantly different than 0-min and the 10-min midpoint across the blocks ($p_{adj} \leq 0.033$; Fig. 5B). Lastly, main effects of *time* ($p = 0.010$) and *block* ($p = 0.048$) were observed for the buttock pain ratings. Similar to the low back, pain ratings for the buttocks were highest at 20-min ($p_{adj} \leq 0.037$; Fig. 5C). Additionally, buttocks pain was higher during the activity guided tasks ($p_{adj} = 0.014$; Fig. 5C) but not different between the traditional and dynamic sitting for computer work. No significant effects of *sex* were observed for buttocks pain ratings ($p \geq 0.119$).

3.6. Productivity

There were no significant main effects or interactions for *block* or *sex* on measures of productivity, including data entry accuracy ($p \geq 0.436$) and data entry speed ($p \geq 0.116$). Accuracy scores were 92.6 ± 7.2% and 92.5 ± 9.0% in traditional and dynamic sitting, respectively. Individuals demonstrated 9.6 ± 2.2 entries/min in traditional sitting and 9.7 ± 2.7 entries/min in dynamic sitting.

4. Discussion

Although lumbar spine kinematics and pain were similar when performing office work in traditional and dynamic sitting, performing activity guided tasks on the dynamic chair increased lumbar spine movement. The first hypothesis was partially supported as dynamic sitting led to changes in pelvis and lumbar spine kinematics primarily in the frontal plane (Figs. 2 and 3; Table 1). The second hypothesis was also supported as buttocks pain was higher during dynamic sitting, but no differences in low back pain existed between traditional and dynamic sitting (Fig. 5). Collectively, these findings indicate that compared to

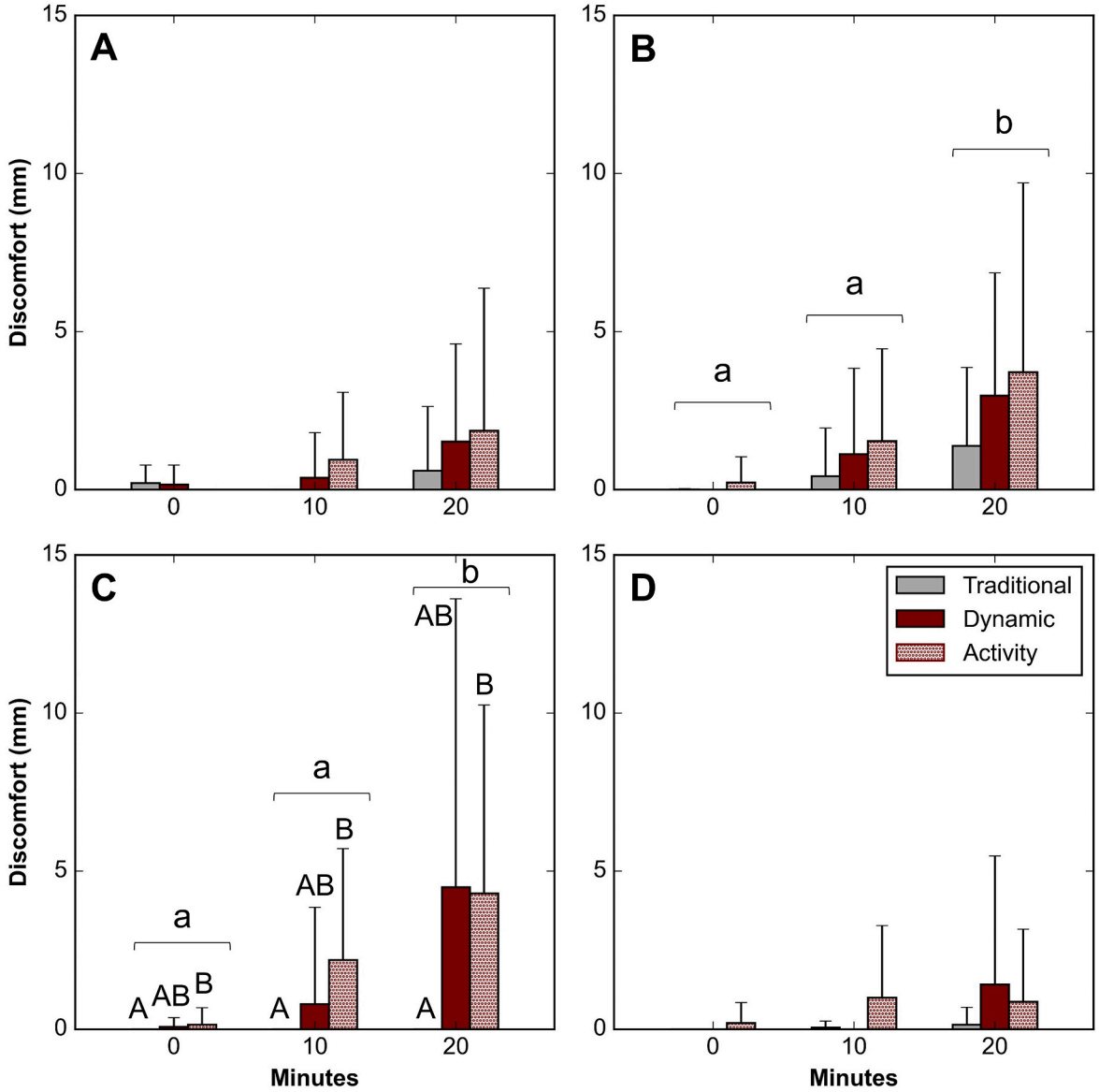


Fig. 5. Mean and (standard deviation) of the pain ratings for the upper back (A), low back (B), buttocks (C), and thighs (D) in each 20-min block. Bars with different lower case letters indicate significant differences between *time*. Bars with different capital letters indicate significant differences between *block*.

sitting in a traditional office chair, dynamic sitting increased lumbar spine movement during the activity guided tasks, without negatively impacting lumbar spine postures, movement, pain, or productivity when performing standardized computer work.

Pelvis and lumbar spine postures and movement were typically not different across the chairs and activities. Dynamic sitting resulted in modest decreases in posterior pelvic tilt, yet no differences in lumbar flexion angles were observed. Mean lumbar flexion was 62 %MaxFlex during simulated office work and 69 %MaxFlex during the activity guided tasks (Fig. 3). These findings are consistent with the seat pan being able to tilt anteriorly (De Carvalho and Callaghan, 2023b) and are similar to lumbar flexion angles previously observed with dynamic sitting on a multiaxial rotating seat pan (Frey et al., 2021). Additionally, the lack of change in lumbar spine angles is consistent with suggestions that the trunk-thigh angle needs to approach 145° in females and 160° in males before there are any corresponding decrease in lumbar flexion (Noguchi et al., 2019). Pelvic obliquity and lumbar lateral bending were not different between the chairs during the seated computer work, wherein mean lateral bending angles of 16 %MaxBend were observed for each of the three blocks (Fig. 4). These angles correspond with lateral angles during seated computer work (Davidson et al., 2024). Only when performing activity guided tasks on the dynamic chair did increases in peak lateral angles for the pelvis and lumbar spine occur, indicating increases in total lateral range of motion during the activity guided tasks (Fig. 4). Taken together, these findings indicate that unless prompted, motion of the pelvis and lumbar spine is similar between traditional and dynamic sitting. Although the dynamic chair provided the opportunity for spine motion in the sagittal and frontal plane, it did not increase lumbar flexion or introduce asymmetrical postures greater than the office chair when performing computer work. The lack of differences in lumbar spine flexion and lateral bending angles may represent a positive finding for this type of dynamic chair as awkward or non-neutral spinal postures are associated with an increased risk of spine injury in ergonomic applications, particularly manual material handling tasks (Marras et al., 1997). Moreover, asymmetrical postures in sitting have been linked to alterations in seat pressure distribution which could contribute to buttocks or low back pain (Al-Eisa et al., 2013; Viggiani et al., 2014). On the other hand, this data also indicates that larger forward and backwards rotation of dynamic chairs may be necessary if the goal of the chair is to decrease flexed spine postures.

Similar to the findings for lumbar spine postures, lumbar spine movement was largely consistent between traditional and dynamic sitting. The number of shifts and fidgets were not different between traditional and dynamic sitting during computer work, but larger lateral fidgets were observed in dynamic sitting (Table 1). Larger shifts and fidgets were also observed during the activity guided tasks compared to the computer work conditions (Table 1). Surprisingly, the number of shifts and fidgets were not different or less during the activity guided tasks; however, this may be related to the thresholds used in calculating these metrics. The shift and fidget dynamic metrics were initially designed to capture movement in a traditional non-rotating seat pan (Dunk and Callaghan, 2010). It is possible that movement in dynamic sitting was not captured by these metrics as the dynamic chair could have facilitated more rapid or smaller movements than those necessary to trigger a fidget or shift. For instance, the fidget movement threshold is based on 3 standard deviations of the movement range and would have been more difficult to achieve on the dynamic chair as there was more variability in participant movement. Greater variability in lumbar movement during the activity guided tasks is supported by the increases in lumber range of motion (Figs. 3 and 4) and large shift and fidget amplitudes (Table 1). Nevertheless, implementing activity guided tasks on a dynamic chair is likely an effective tool for increasing movement at the workplace. Similar to recommendations to rotate between sitting and standing and taking breaks throughout the workday, the current findings indicate that performing activities on a dynamic chair encourages movement of the spine. Moreover, many individuals report low back pain with prolonged standing (Gallagher et al., 2011; Nelson-Wong and Callaghan, 2014), while the current findings, along with other analyses have not shown significant increases in low back pain development with dynamic sitting (Frey et al., 2021; Léger et al., 2022; O'Sullivan et al., 2012).

Indeed, pain of the upper back, low back, and thighs remained low in both traditional and dynamic sitting, but buttocks pain was highest for dynamic sitting. Specifically, 6 of 16 participants reported meaningful increases (i.e., >~10 mm (Gallagher et al., 2014; Gallagher and Callaghan, 2015; Hägg et al., 2003; Nelson-Wong and Callaghan, 2010; Schinkel-Ivy et al., 2013)) in the buttocks (3 females; 1 in Dynamic, 1 in Activity, and 1 in both) or low back (3 males; 1 in Dynamic, 1 in Activity, and 1 in both) during dynamic sitting. Higher reports of buttocks pain were observed for dynamic sitting in general, but significant differences were only observed between traditional sitting and the activity guided tasks (Fig. 5). Buttocks pain in the activity was likely because participants were required to apply force to the seat pan to move their pelvis. Additionally, increases in vertical and anterior shear forces in dynamic sitting, particularly during the activity guided tasks (Table 2), also indicate increases in lower limb involvement to resist sliding forward (Noguchi et al., 2019). Subsequent friction from the seat pan could have also contributed to buttocks pain. Pain may also be higher during the activity guided tasks as they were always completed last (i.e., a cumulative total of 40-min on the dynamic chair with only short breaks between) and participants were required to complete the activities for 20-min to ensure time equalization between the three blocks. In practice, the activity guided tasks would be recommended over shorter durations (<10 min) to facilitate training individuals how to use the dynamic chair, for micro or short breaks during the workday, or when performing non-visual tasks like taking a phone call. Completing activity guided tasks at the workstation are not meant to replace active breaks (e.g., going for a walk), but rather provide additional opportunities for movement at the workplace. Further, implementing accommodation training when starting to use a dynamic chair may be effective for mitigating pain over time (Jackson et al., 2013).

There are limitations to be acknowledged. First, the activity guided tasks were designed to elicit motion of the pelvis on a dynamic chair and are not possible on a traditional office chair. Without the activity guided tasks on a traditional office chair, a factorial design (i.e., *chair* × *task*) was not possible. Additionally, the applicability of the current findings are limited as a dynamic chair, coupled to a phone, is required to complete these specific activity guided tasks. Second, we assessed 20-min blocks, which is well below what would be typical of an entire workday or would be likely to occur during a single bout of sitting before a break (Matthews et al., 2008; Parry and Straker, 2013). It is possible that with more prolonged bouts of seated work, pain ratings could continue to rise (De Carvalho and Callaghan, 2023a; Gallagher and Callaghan, 2015). However, as indicated by values of approximately 0 mm at the start of each block (Fig. 5), pain recovered quickly to those at the start of the collection during the short transitions between the 20-min blocks. Lastly, only young and healthy participants, with a small variation in anthropometry were examined. The homogeneity amongst participants may have contributed to the lack of significant differences which existed between males and females.

Overall, dynamic sitting, facilitated by a multiaxial rotating seat pan, while performing computer work did not result in large changes to pelvis and lumbar postures and movement or work productivity. Although upper back, low back, and thigh pain remained low during both traditional and dynamic sitting, buttocks pain tended to be higher during dynamic sitting, particularly in the activity guided tasks. The current findings indicate that introducing forward/backwards and lateral movement of the seat pan can occur without negatively impacting low back kinematics, pain, or productivity. Further, a dynamic chair facilitates the performance of activity guided tasks which appear to be an effective means of increasing spine movement at a user's workstation. However, accommodation training and performing the activity guided

tasks in bouts of less than 10 min may be required to mitigate pain responses.

CRediT authorship contribution statement

Jessa M. Davidson: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Jackie D. Zehr:** Conceptualization, Formal analysis, Investigation, Methodology, Writing – review & editing. **Paolo B. Dominelli:** Data curation, Investigation, Supervision, Writing – review & editing. **Jack P. Callaghan:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Writing – review & editing.

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